Project Ideation & Design Phase
Problem - Solution Fit Canvas

Date	31 January 2026
Team ID	anshrathore0202-cyber
Project Name	F1 Racing Intelligence Analysis
Maximum Marks	4 Marks

Problem - Solution Fit Overview:

The Problem-Solution Fit Canvas ensures that the proposed Business Intelligence solution directly addresses real user pain points regarding Formula 1 data accessibility, processing complexity, and visual analytics.

PROBLEM DOMAIN

1. Target Customer Segment

- **Motorsports Data Analysts:** Need structured multi-season F1 telemetry for performance modeling.
- **Formula 1 Enthusiasts & Journalists:** Seeking intuitive visual stories behind lap times and race strategies.
- **Strategy Researchers:** Requiring quantitative stint pace & pit-stop efficiency data.

2. Key Customer Problems

- **Data Fragmentation:** Race telemetry is scattered across raw API endpoints, static PDFs, and unstructured CSV files.
- **Time-Consuming Wrangling:** Analysts spend hours cleaning null values and formatting timestamps before analysis.
- **Lack of Interactive Visuals:** Static tables fail to reveal qualifying vs race finish position dynamics dynamically.

3. Existing Alternatives & Disadvantages

- **Raw CSV Spreadsheets:** Cluttered, prone to manual errors, and lacking real-time filtering capabilities.
- **Basic Broadcast Stats:** Limited to real-time race broadcasts without deep multi-season historical comparison tools.

SOLUTION DOMAIN

1. Proposed BI Solution

- **Unified Data Pipeline:** Automated Python ETL scripts ingesting and structuring multi-season Ergast F1 telemetry.
- **Interactive Tableau Dashboards:** Rich visual tools enabling head-to-head driver stint pace and constructor score analysis.
- **Custom Metric Calculations:** Automated computation of pit-stop efficiency and qualifying delta.

2. Value Proposition

- **Instant Insights:** Reduces data exploration time from hours to seconds with interactive visual filters.
- **Data Accuracy & Cleanliness:** Standardized schemas ensure reliable historical comparisons across eras.
- **Engaging Storytelling:** Transforms complex lap data into clear graphic representations for all user levels.

3. Solution Fit & Verification Metrics

- **Query Speed:** Visual filter responses load under 2 seconds across 100,000+ telemetry rows.
- **User Adoption:** Seamless dashboard exploration across season, circuit, driver, and weather parameters.